

HanOmegaNegotiator: Robust Adaptive Negotiation for the HAN 2026 League

Team 507

Abstract

HanOmegaNegotiator is a non-LLM bilateral negotiation agent developed for the Human-Agent Negotiation (HAN) 2026 League. It combines a time-dependent aspiration policy, rational-outcome search, frequency-based opponent modeling, counterfactual acceptance, and concise natural-language messages. The final agent adds two mechanisms aimed at the lower tail of negotiation performance. First, it detects opponents that cycle through previously used offers and responds with high-scoring unseen proposals, preventing reciprocal-concession strategies from stalling. Second, it uses a targeted ordinal anchor search after repeated textual rejection. In a deterministic 456-match benchmark covering 19 scenarios, 12 opponent families, and both negotiation roles, the final agent achieved 0.9978 agreement rate and 0.4651 average normalized utility. The first quartile improved from 0.2402 to 0.2916 without reducing performance against non-cycling time-based opponents.

1 Motivation

HAN uses the alternating offers protocol augmented with text communication. An effective agent must protect its reservation value, adapt to heterogeneous partners, and produce understandable messages for human negotiators. The final online mock tournament contained 24 competitors and 576 negotiations. The submitted agent ranked 18th with a mean score of 4,233, a median of 2,500, and a first quartile of zero. It produced no negotiator exceptions, indicating that the main weakness was negotiation quality in difficult matches rather than runtime stability.

This result motivated a lower-tail objective. Average utility alone can hide a strategy that performs well in easy negotiations but receives no advantage in many difficult ones. Development therefore tracked agreement rate, average normalized utility, first-quartile utility, and the mean of the worst quartile.

2 Architecture

At initialization, the agent enumerates the discrete outcome space, evaluates every outcome with its private utility function, and sorts outcomes by utility. Outcomes below the reservation value are excluded from the rational set. During each turn the agent:

1. observes the partner’s offer and optional text;
2. updates value frequencies, issue stability, offer repetition, and the best received offer;
3. applies the acceptance policy;
4. constructs and scores a frontier of rational counteroffers;
5. returns a counteroffer with a short contextual message.

Expensive global ranking is performed once. Online decisions operate on a bounded candidate frontier, keeping the strategy suitable for interactive use.

3 Offer Generation and Concession

The base aspiration threshold is

$$A(t) = r + (u_{\max} - r)(1 - t^e), \quad (1)$$

where r is the reservation value, $u_{\max}$ is the best attainable utility, t is relative time, and $e = 4.9$ is the concession exponent. The threshold is adjusted by observed stagnation and the best offer received so far. Near the deadline it is capped to permit rational agreement.

The candidate frontier combines:

- high-utility outcomes above the current threshold;
- outcomes close to the partner’s current offer;
- local repairs that recover self-utility with low estimated opponent loss;
- domain-specific trade and allocation candidates;
- incremental variants that change one issue toward the agent’s preferred value;
- late rational outcomes with high estimated acceptance likelihood.

Candidate scoring combines normalized self-utility, estimated partner utility, acceptance likelihood, concession efficiency, fairness, closeness to the latest partner offer, and novelty. Self-utility dominates early. Opponent acceptability and offer closeness receive greater weight after relative time 0.65 and against harder partners.

4 Opponent Modeling and Cycle-Aware Exploration

The ordinary opponent model is frequency based. For each issue, the agent tracks offered-value counts, entropy, stability, and change frequency. Repeated values are interpreted as preferred values, while stable issues receive greater weight. This lightweight model is deliberately used for ranking rather than treated as an exact reconstruction of the hidden utility function.

Failure-trace analysis found an important protocol interaction. Some reciprocal strategies, especially MiCRO-like agents, concede when they receive a new offer. HanOmegaNegotiator could alternate between two good previous offers, causing the partner to stop conceding. The final policy records all sent proposals and computes the partner-offer uniqueness ratio

$$\rho = \frac{|\{o_1, \dots, o_n\}|}{n}. \quad (2)$$

After at least six observations and relative time 0.20, a partner is classified as cycling when $\rho \leq 0.75$. If the normal policy selects an already sent offer against such a partner, the agent instead sends the highest-scoring unseen outcome on the current rational frontier. The mechanism is inactive for non-cycling partners, preserving the established Boulware, linear, and conceder behavior.

5 Acceptance and Text-Aware Recovery

Offers below the reservation value are always rejected. An offer is accepted if it exceeds the aspiration threshold or is sufficiently close to the next planned counteroffer. Acceptance slack is expressed as 5–10% of the feasible utility span rather than as a raw utility constant. This makes acceptance behavior consistent across utility scales such as $[0, 1]$ and $[0, 100]$.

Late rules accept strong repeated offers, offers close to the best fallback, and offers that dominate the likely next counter from the partner’s perspective. A separate text-anchor recovery mechanism activates only after relative time 0.75 when the partner repeatedly sends a textual offer that remains below the agent’s reservation value. For numeric issues, it estimates preference by distance from modal values and searches for a rational compromise. Rejected probes are diversified by their changed-issue signature, allowing the agent to try a different compromise axis instead of changing the same issue repeatedly.

Messages are template based and do not require an external model. They acknowledge fairness, urgency, or stated needs when detected, summarize at most two material changes, and avoid revealing the private utility function.

6 Evaluation

Evaluation used the official HAN template scenarios and generated Grocery, Island, Trade, and generic multi-issue domains. Twelve opponent families covered BOA, Boulware, linear, conceder, Pareto-following, MiCRO, Tit-for-Tat, TopFraction, tough, simple, and template-adaptor strategies. Every scenario was tested with the agent in both roles under deterministic Python and NumPy seeds.

Version	Agreement	Mean norm.	Q1 norm.	Bottom-quartile mean
Pre-final agent	0.9912	0.4499	0.2402	0.0517
Final agent	0.9978	0.4651	0.2916	0.0741

Table 1: Primary deterministic benchmark over 456 negotiations at 40 steps.

Cycle-aware exploration increased MiCRO normalized utility from 0.2937 to 0.4007 and Tit-for-Tat utility from 0.4626 to 0.5375. Scores against BOA, Boulware, conceder, linear, and template-adaptor families were unchanged. At 20 steps, mean normalized utility improved from 0.4258 to 0.4363 and Q1 from 0.1958 to 0.2148. At 80 steps, mean utility improved from 0.4422 to 0.4617 and Q1 from 0.2239 to 0.2602.

An independent holdout used unused generated and random seeds for another 456 matches. The final agent improved mean normalized utility from 0.5144 to 0.5285, Q1 from 0.3260 to 0.3540, and agreement rate from 0.9978 to 1.0000. This result indicates that the retained improvement was not specific to the tuning seeds.

7 Limitations and Conclusion

The strategy assumes a discrete outcome space that can be enumerated. Its frequency model may be inaccurate when a partner deliberately varies offers or when issue values have complex non-monotonic semantics. Cycle detection is therefore conservative and affects only offer novelty; it does not override the reservation value or rational frontier.

HanOmegaNegotiator combines ambitious early bidding, scale-invariant acceptance, lightweight opponent modeling, human-readable messages, and targeted recovery from repeated or cycling behavior. Benchmark and holdout results show that the final design improves agreement, average utility, and lower-tail robustness while preserving performance against already strong non-cycling opponent families.